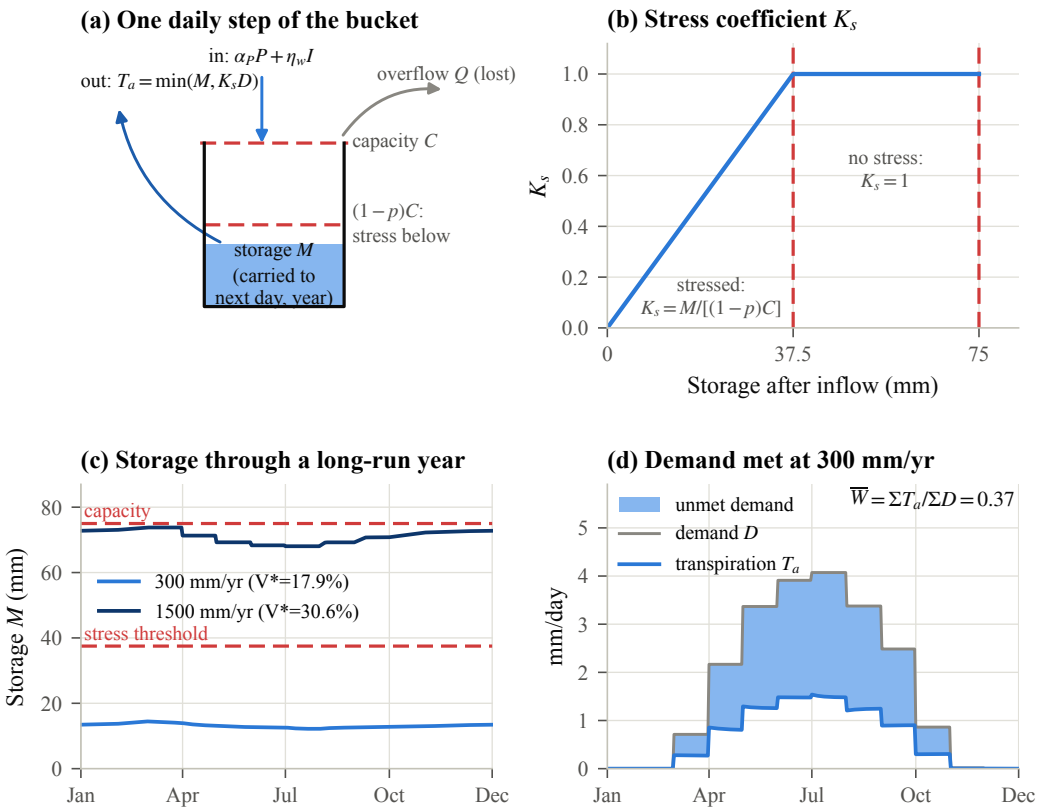

(a) One daily step of the bucket
in: $\alpha_P P + \eta_w I$
out: $T_a = \min(M, K_s D)$
overflow Q (lost)
capacity C
$(1-p)C$:
stress below
storage M
(carried to
next day, year)
(b) Stress coefficient K_s
K_s
1.0
0.8
0.6
0.4
0.2
0.0
no stress:
$K_s = 1$
stressed:
$K_s = M/[(1-p)C]$
0
37.5
75
Storage after inflow (mm)
(c) Storage through a long-run year
Storage M (mm)
80
60
40
20
0
capacity
300 mm/yr (V*=17.9%)
1500 mm/yr (V*=30.6%)
stress threshold
Jan
Apr
Jul
Oct
Dec
(d) Demand met at 300 mm/yr
$\overline{W} = \Sigma T_a / \Sigma D = 0.37$
unmet demand
demand D
transpiration T_a
mm/day
5
4
3
2
1
0
Jan
Apr
Jul
Oct
Dec